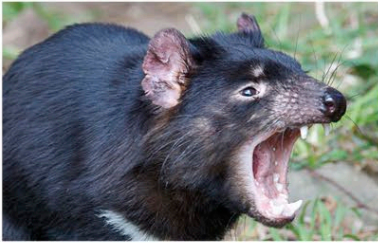


MAMMALS

The mammals are very successful animals. They are able to occupy almost every terrestrial habitat, and some visit the deep oceans between breaths of air. However, this has not always been the case; at 210 million years old, mammals are a relatively recent group.

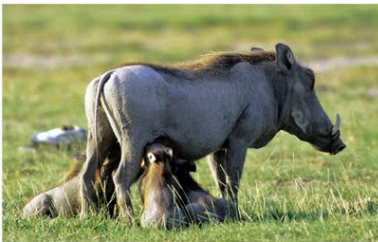
PHYLUM	CHORDATA
CLASS	MAMMALIA
ORDERS	28
FAMILIES	160
SPECIES	About 6,300



A single jawbone articulated directly to the skull gives mammals, such as this Tasmanian devil, a powerful bite.



Baleen is made of a protein called keratin, and baleen whales use it to strain sea water, trapping food in their mouth.



By sucking milk from their mother, young warthogs obtain all the nourishment they need during the first weeks of life.

DEBATE

EQUALLY WELL ADAPTED

Marsupial features are sometimes described as primitive. This implies that marsupials, whose young are born in an embryonic state and develop in a pouch, are less advanced than placental mammals, which bear conventional young. In fact, the two groups emerged at the same time about 175 million years ago.

Mammals owe their success to a unique combination of adaptations that allowed them to replace reptiles as the dominant form of animal life on Earth. Like reptiles, mammals breathe air, but unlike their scaly ancestors, they are warm-blooded, so they are able to burn fuel (food) in order to maintain a constant, warm body temperature. This creates conditions in which the internal chemical processes that sustain life can happen most efficiently, without relying directly on heat from the sun. Uniquely, mammals have hair, which reduces heat loss and allows an animal to be active in cold climates and at night. Because fur can be moulted and regrown, it is seasonally adjustable too.

VARIATIONS ON A THEME

The basic mammal skeleton is strong, with upright limbs supporting the body from below. This arrangement allows land-dwelling mammals to walk, run, and leap, but the basic structure is highly adaptable, and has been modified for swimming, as in seals and cetaceans; for flying, as in bats; and for climbing or swinging through the trees, as in some primates. The mammalian skull has powerful jaws, with a single lower jawbone articulating directly with the skull, and a diverse range of teeth adapted to a huge variety of food types. Certain other bones, which in reptiles contribute to the lower jaw, are put to a new use by mammals – they have become the three tiny bones of the inner ear, allowing for greatly enhanced hearing. The mammalian skull also serves to protect the brain, which is larger in mammals than in other groups, hinting at increased processing power. With a big brain comes intelligence, and an unrivalled ability to learn, to remember, and to engage in complex behaviour.

These abilities take time to fine-tune, and young mammals do so during the extended period of parental care that begins while they are being fed on milk produced by their mother's mammary glands. These unique organs evolved from sebaceous glands, which originally provided secretions to condition the skin and perhaps to prevent eggs drying out, and it is for them that the entire group is named.

FUR AS INSULATION >

The air trapped in a sea otter's dense fur insulates it from the cold water in which it lives.

